

SFTP Deployment Guide

Deployment Guide — Enterprise SFTP Management System

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Deploy the full stack — atmaz/sftp container + PostgreSQL + Go REST API (+ web admin) — in **rootless Podman** with podman-compose, driven by systemd --user units. No root, no sudo, no Docker daemon (\$11.4.161).

Sections marked **PLANNED — lands with STREAM-x** describe pieces not yet built (see docs/Issues.md); commands there are the committed design, not yet executable. Everything else is executable today.

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1. Prerequisites

- Linux host, Podman ≥ 3.4 installed (host reference: Podman 5.7.1, 64 GB RAM).
- podman-compose — “a script to run docker-compose.yml using podman”, daemon-less, runs as a regular user:

```
pip3 install --user podman-compose
command -v podman-compose # must resolve
```

- No root needed at any step. Rootless mode is Podman's unprivileged-user operation mode (user namespaces); every command in this guide runs as the service user.
- Host ports free: 7721/tcp (SFTP), 7722/tcp (REST API). Check:

```
ss -tlnp | grep -E ':(7721|7722) ' || echo "ports free"
```

- sftp client for verification (openssh-clients).

2. Environment setup

```
git clone <repo-url> sftp && cd sftp
cp .env.example .env
# .env.example lands with STREAM-1 (ATM-001)
chmod 600 .env # credentials file – §11.4.10
```

Key variables (full surface in .env.example):

Variable	Default	Purpose
SFTP_PORT	7721	Host port → container sshd :22
API_PORT	7722	REST API bind port
SFTP_DATA_DIR	./data	Host dir for all user homes
DB_DRIVER	sqlite	sqlite (dev) / postgres (prod)
SUPER_ADMIN_USER	—	Bootstrap super-admin name (setup.sh prompts)

Never commit .env (git-ignored, §11.4.30). Secrets are generated by scripts/setup.sh and never echoed (§11.4.10).

3. Compose stack

PLANNED — lands with STREAM-1 (ATM-001): deploy/docker-compose.yml. Shape (validated via `podman-compose -f deploy/docker-compose.yml config`):

```
services:
  sftp:
    image: atmoz/sftp:latest
    ports:
      - "${SFTP_PORT:-7721}:22"
```

```

volumes:
  - ${SFTP_DATA_DIR:-./data}:/home
  - ./users.conf:/etc/sftp/users.conf:ro
  - ./ssh_host_keys:/etc/ssh/host_keys # persistent host
    keys – no MITM churn on recreate
restart: unless-stopped
postgres:
  image: postgres:16-alpine
  environment: { POSTGRES_DB: sftp, POSTGRES_USER: sftp,
    POSTGRES_PASSWORD_FILE: /run/secrets/db_password }
  volumes: [ "pgdata:/var/lib/postgresql/data" ]
api:
  build: ./api
  ports: [ "${API_PORT:-7722}:7722" ]
  depends_on: [ postgres, sftp ]
  env_file: [ .env ]
volumes: { pgdata: {} }

```

Notes: users.conf is mounted read-only and **rendered atomically by the API** (ATM-002/003) — never hand-edited in production. Container sshd always listens on internal 22; the host port is the mapping. atm0z/sftp defaults users to plaintext passwords in users.conf; the API instead writes encrypted hashes (:e marker, e.g. user:\$1\$...:e:uid:gid) — see [security_guide.md](#).

4. First run — setup.sh

PLANNED — lands with STREAM-6 (ATM-006):

```
bash scripts/setup.sh
```

The script (design): validates .env, generates secrets into git-ignored files, creates the super-admin via the API bootstrap endpoint, runs a smoke probe, prints only non-secret status lines. It is idempotent — safe to re-run; existing secrets are never overwritten.

5. systemd -user service install

PLANNED — lands with STREAM-6 (ATM-006): units under deploy/systemd/, installed to ~/.config/systemd/user/:

```

bash scripts/service_ctl.sh install # copies units, daemon-
    reload, enable --now
bash scripts/service_ctl.sh status

```

Units are --user scoped (no sudo). To survive logout, enable linger **once** (host policy permitting — this is the only step that may need an administrator on managed hosts):

```
loginctl enable-linger "$USER"  
# see troubleshooting_guide.md §6 if unavailable
```

6. Verification

Run after every install/upgrade (full automation: tests/ suites per ATM-009):

```
# 1. Containers healthy  
podman ps --format '{{.Names}} {{.Status}}'  
  
# 2. API health (PLANNED — lands with STREAM-2 / ATM-002)  
curl -fsS http://127.0.0.1:${API_PORT:-7722}/healthz # expect  
{"status":"ok"}  
curl -fsS http://127.0.0.1:${API_PORT:-7722}/readyz  
  
# 3. SFTP login round-trip with a read_write test account  
sftp -P ${SFTP_PORT:-7721} <user>@127.0.0.1 <<'EOF'  
put /etc/hostname uploaded.txt  
ls -la  
bye  
EOF  
  
# 4. Audit log shows the session (PLANNED — ATM-002 audit trail)  
curl -fsS -H "Authorization: Bearer $TOKEN" http://127.0.0.1:$  
{API_PORT:-7722}/api/v1/audit | head
```

A PASS for each step requires the positive output shown — “no error printed” alone is not proof (§11.4 anti-bluff).

7. Upgrade

1. git fetch --all --prune + ff-merge (§11.4.37/§11.4.113 — never force-push).
2. bash scripts/backup.sh — snapshot DB + data/ + users.conf (PLANNED — ATM-006).
3. Pull new images: podman-compose -f deploy/docker-compose.yml pull.
4. bash scripts/service_ctl.sh restart.
5. DB migrations run automatically on API start (embedded migrations, ATM-002); watch podman logs <api-container>.
6. Re-run §6 Verification.

8. Rollback

1. bash scripts/service_ctl.sh stop.
2. Restore the previous docker-compose.yml/image tag from git, or podman tag-pin the prior image.

3. Restore the DB snapshot: `bash scripts/backup.sh restore <snapshot-id>` (PLANNED — ATM-006).
4. `bash scripts/service_ctl.sh start` + full §6 verification.
5. Record the rollback in `docs/qa/<run-id>/` with the captured logs (§11.4.83).

9. Related docs

- [user_management_guide.md](#) — accounts, permissions, keys
 - [security_guide.md](#) — hardening, secrets, threat model
 - [troubleshooting_guide.md](#) — failure-mode playbooks
 - [../tutorials/quickstart.md](#) — 10-minute first deployment
 - `docs/research/mvp/MVP.md` — baseline atmaz/sftp compose reference
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Sources verified (2026-07-11)

- Internal: `docs/research/mvp/MVP.md` (baseline compose, `users.conf` format, chroot notes) — `docs/Issues.md` ATM-001/002/003/006 (stack design) — `docs/CONTINUATION.md` (host facts: Podman 5.7.1 rootless, Go 1.26.2).
- External fetched this revision: <https://github.com/atmoz/sftp> (`users.conf` grammar `user:pass[:e][:uid[:gid[:dir...]]], :e` encrypted marker, `.ssh/keys/` key mount, chroot home not user-writable, port 22 internal) · <https://github.com/containers/podman-compose> (“a script to run docker-compose.yml using podman”, rootless `pip --user install`) · <https://docs.podman.io/en/latest/> (rootless = non-privileged user operation).
- To re-verify before each release (§11.4.99): atmaz/sftp README (image behavior), podman-compose README (install/compat), Podman rootless tutorial pages.